

Supplementary Computational Material

Frozen-Jacobian methods with predictor and memory: order, optimal selection, and fusion

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This supplement provides detailed numerical breakdowns and secondary graphical representations supporting the computational experiments reported in the accompanying article. Full machine-readable datasets and the computational code are provided in the repository. All timing values reported below are archived measurements obtained under the experimental protocols described in the article.

1 Detailed external-comparator timings

Tables S1 and S2 report the detailed results at $\text{TOL} = 10^{-12}$. The article summarizes all nine configurations per field. Full-precision CSV files, including the other tolerances, are supplied in `data/reference/p5_m6_results.csv` and `p7_m8_results.csv`. The relative quantities are $\Delta t = 100(t_A/t_B - 1)$ and $\Delta W = 100(W_A/W_B - 1)$ for $\kappa = 1$; negative values favor method *A*. Parentheses contain the IQR, in the same units as the median. The time separation rule is $|\tilde{t}_A - \tilde{t}_B| > 2(\text{IQR}_A + \text{IQR}_B)$; it is descriptive, not a formal statistical test.

Table S1: Comparison P_5/M_6 at $\text{TOL} = 10^{-12}$. Times and IQRs are in milliseconds.

Field	δ	Cycles	P_5 time (IQR)	M_6 time (IQR)	Δt [%]	ΔW [%]
H_1	0.10	1/1	0.1258 (0.0064)	0.1252 (0.0075)	+0.48	+6.78
H_1	0.30	2/2	0.2478 (0.0107)	0.2428 (0.0175)	+2.06	+8.89
H_1	0.60	2/2	0.2495 (0.0151)	0.2423 (0.0124)	+2.97	+8.89
H_2	0.10	1/1	0.1752 (0.0104)	0.1747 (0.0124)	+0.29	+5.91
H_2	0.30	1/1	0.1730 (0.0136)	0.1731 (0.0141)	-0.06	+5.91
H_2	0.60	2/2	0.3514 (0.0263)	0.3362 (0.0186)	+4.52	+7.37
H_3	0.10	1/1	0.2442 (0.0389)	0.2506 (0.0310)	-2.55	+3.97
H_3	0.30	1/1	0.2411 (0.0127)	0.2499 (0.0207)	-3.52	+3.97
H_3	0.60	2/2	0.4778 (0.0132)	0.4909 (0.0275)	-2.67	+4.98
H_4	0.10	1/1	0.5495 (0.0301)	0.5704 (0.0699)	-3.66	+3.08
H_4	0.30	1/1	0.5505 (0.0356)	0.5772 (0.0552)	-4.63	+3.08
H_4	0.60	2/2	1.1382 (0.1175)	1.1402 (0.1064)	-0.18	+3.84
H_5	0.10	1/1	1.0378 (0.0843)	1.0821 (0.1272)	-4.09	+2.44
H_5	0.30	1/1	1.0760 (0.0742)	1.1164 (0.0804)	-3.62	+2.44
H_5	0.60	2/2	2.0588 (0.1305)	2.1217 (0.2128)	-2.96	+3.05

Table S2: Comparison P_7/M_8 at $\text{TOL} = 10^{-12}$. Times and IQRs are in milliseconds.

Field	δ	Cycles	P_7 time (IQR)	M_8 time (IQR)	Δt [%]	ΔW [%]
H_1	0.10	1/1	0.1537 (0.0084)	0.1202 (0.0050)	+27.87	+15.71
H_1	0.30	1/1	0.1558 (0.0102)	0.1220 (0.0108)	+27.70	+15.71
H_1	0.60	2/2	0.3057 (0.0103)	0.2274 (0.0088)	+34.43	+17.94
H_2	0.10	1/1	0.2047 (0.0096)	0.1774 (0.0155)	+15.39	-5.07
H_2	0.30	1/1	0.2128 (0.0139)	0.1808 (0.0169)	+17.70	-5.07
H_2	0.60	1/1	0.2103 (0.0115)	0.1801 (0.0106)	+16.77	-5.07
H_3	0.10	1/1	0.2850 (0.0176)	0.2844 (0.0148)	+0.21	-23.53
H_3	0.30	1/1	0.2873 (0.0162)	0.2856 (0.0116)	+0.60	-23.53
H_3	0.60	1/1	0.2869 (0.0123)	0.2881 (0.0168)	-0.42	-23.53
H_4	0.10	1/1	0.5839 (0.0392)	0.7361 (0.0495)	-20.68	-30.59
H_4	0.30	1/1	0.6107 (0.0744)	0.7729 (0.0663)	-20.99	-30.59
H_4	0.60	1/1	0.5947 (0.0471)	0.7348 (0.1047)	-19.07	-30.59
H_5	0.10	1/1	1.1568 (0.0984)	1.7986 (0.1976)	-35.68	-34.90
H_5	0.30	1/1	1.0842 (0.0818)	1.6504 (0.1198)	-34.31	-34.90
H_5	0.60	1/1	1.0758 (0.0352)	1.6450 (0.1357)	-34.60	-34.90

2 Full sensitivity of the optimal members

Table S3 reports all six control points. The article displays two representative endpoints. Stability throughout $[1, 4.5]$ follows from affine normalized-cost differences and the unimodality argument, not from this finite list alone.

Table S3: Stationary family optima and 0.5% near-optimal bands.

Field	κ	N_P^*	m_S^*	$10^6 \eta_P^*$	$10^6 \eta_S^*$	Advantage [%]	P/S bands
H_1	1	3	3	172.461715	165.035043	4.50	$\{3\}/\{3\}$
H_1	1.2	3	3	171.409882	164.058504	4.48	$\{3\}/\{3\}$
H_1	1.298	3	3	170.899153	163.584207	4.47	$\{3\}/\{3\}$
H_1	1.7	3	3	168.835587	161.666981	4.43	$\{3\}/\{3\}$
H_1	3.23511	3	3	161.393776	154.741466	4.30	$\{3\}/\{3\}$
H_1	4.45444	3	3	155.934456	149.649474	4.20	$\{3\}/\{3\}$
H_5	1	9	10	0.555516	0.543469	2.22	$\{8, 9, 10, 11\}/\{9, 10, 11\}$
H_5	1.2	9	10	0.554944	0.542925	2.21	$\{8, 9, 10, 11\}/\{9, 10, 11\}$
H_5	1.298	9	10	0.554664	0.542659	2.21	$\{8, 9, 10, 11\}/\{9, 10, 11\}$
H_5	1.7	9	10	0.553518	0.541570	2.21	$\{8, 9, 10, 11\}/\{9, 10, 11\}$
H_5	3.23511	9	10	0.549186	0.537452	2.18	$\{8, 9, 10, 11\}/\{9, 10, 11, 12\}$
H_5	4.45444	9	10	0.545794	0.534226	2.17	$\{8, 9, 10, 11\}/\{9, 10, 11, 12\}$

3 Secondary graphical representations

Figures S1 and S2 provide graphical views of the secondary comparisons. The underlying measurements are the same as those reported in the data files. A configuration range is not an IQR or confidence interval. No interpolation establishes a universal time crossover or monotonicity.

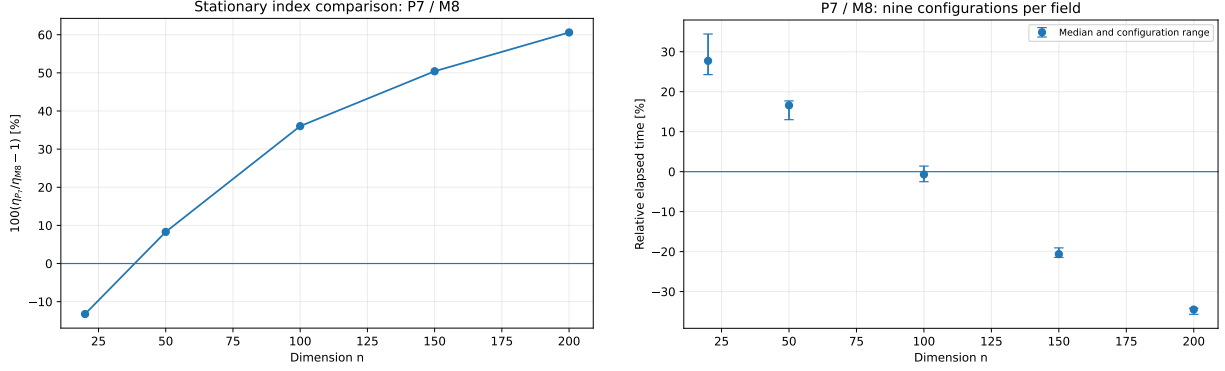


Figure S1: P_7/M_8 : stationary index advantage (left) and median with range of relative elapsed time over nine configurations per field (right). Positive index differences and negative time differences favor P_7 . Time-separation decisions are made separately for each configuration.

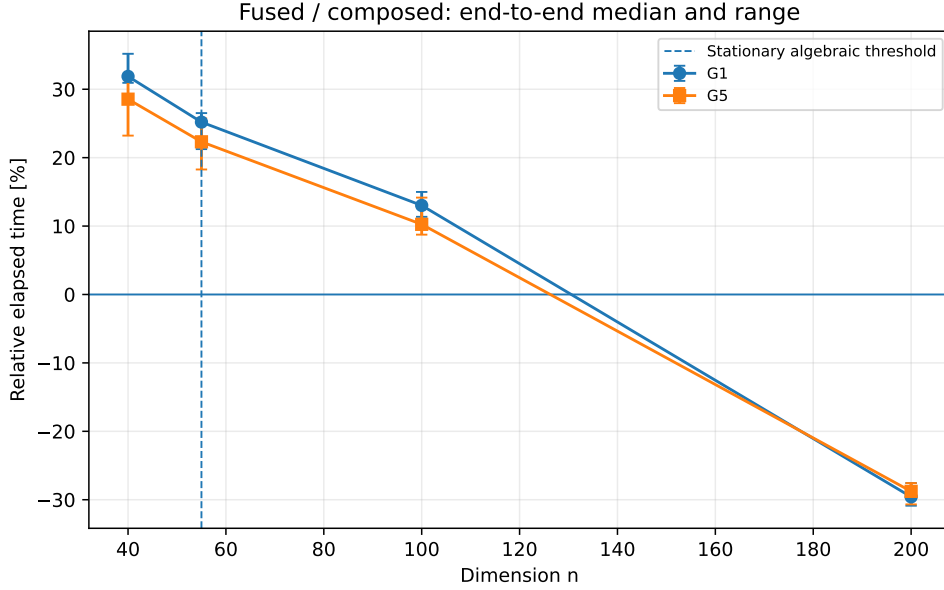


Figure S2: End-to-end comparison of $\hat{P}_2$ with $P_2 \circ P_2$: median and range of relative elapsed time. The line $n = 55$ is the stationary algebraic threshold, not an estimated temporal crossover. Negative values favor fusion.

Data provenance. For G_5 , the archived fusion work columns were computed with the evaluation-cost convention stored with the original runs; this is documented in `docs/KNOWN_ISSUES.md`. The repository also provides work values recomputed with the canonical cost convention used in the article. The elapsed-time values in Figure S2 are unaffected.